Project Workflow: Week 5

Phase: Procedural Island & Player Movement (Treasure Hunt Game)

1. Objective

The primary goal of Week 5 was to implement dynamic procedural island generation and full player movement mechanics with state handling for the Treasure Hunt game. Building upon the Week 4 game window and main loop, this phase focused on generating a grid-based tile map, tracking player position and states, and wiring up user controls via keyboard listeners.

2. Detailed Task Breakdown

- **Procedural Island Generation (Map.java):** Designed an algorithmic grid map that procedurally generates terrain tiles (water borders, sand shores, and island grass center) to create a randomized, natural-looking island environment.
- **Player State & Movement Mechanics (Player.java):** Developed player entity representation, including grid position tracking (X, Y coordinates), direction facing, and boundary checking to prevent moving onto water tiles.
- **Keyboard Control Wiring:** Integrated KeyListener to map WASD and Arrow Key inputs to player movement actions with smooth updates inside the game loop.
- **Graphics Rendering & Visual Integration:** Extended JPanel rendering logic to draw both procedural tiles and the animated player character dynamically on screen.

3. Deliverables

- **Procedural Island Generator (Map.java):** A working map class capable of generating grid-based island terrain layouts.
- **Player Controller (Player.java):** Player entity class managing movement logic, direction, and grid position states.
- **Interactive Control System:** Fully wired KeyListener handling WASD and Arrow key inputs.
- **Visual Tile & Entity Rendering:** Updated game panel rendering both the island terrain and player entity visually in real-time.

4. Tools & Resources

- **Development Environment:** Java JDK 11+, VS Code / IntelliJ IDEA
- **Libraries & Technologies:** Java Swing (JFrame, JPanel), Java AWT (Graphics, Graphics2D), Java Event Handling (KeyListener, KeyEvent)